



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Computer Networks Lab Project

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Enterprise-Level Network Design and Implementation Using Cisco Packet Tracer

1. Introduction

In today's global environment, computer networks form the backbone of communication and data sharing between geographically distributed locations. Organizations rely on enterprise-level networks to ensure secure, reliable, and scalable connectivity between multiple sites. These networks integrate Local Area Networks (LANs) with Wide Area Networks (WANs) using routing protocols and network services.

This project focuses on designing and implementing an **enterprise-level network infrastructure** that connects multiple independent networks (Islamabad, Lahore, and Karachi) using Cisco Packet Tracer. The network integrates VLANs, routing, DHCP, DNS, and WAN technologies such as PPP and RIP version 2.

2. Project Objectives

The main objectives of this project are:

- To design a multi-site enterprise network.
- To implement VLANs for logical network segmentation.
- To configure inter-VLAN routing using router-on-a-stick.
- To deploy WAN connectivity using serial links and PPP encapsulation.
- To implement RIP version 2 for dynamic routing.
- To configure DHCP, DNS, and Web servers.
- To test network functionality using ping, tracer, and DNS resolution.

3. Network Topology Overview

The enterprise network consists of **three main sites**:

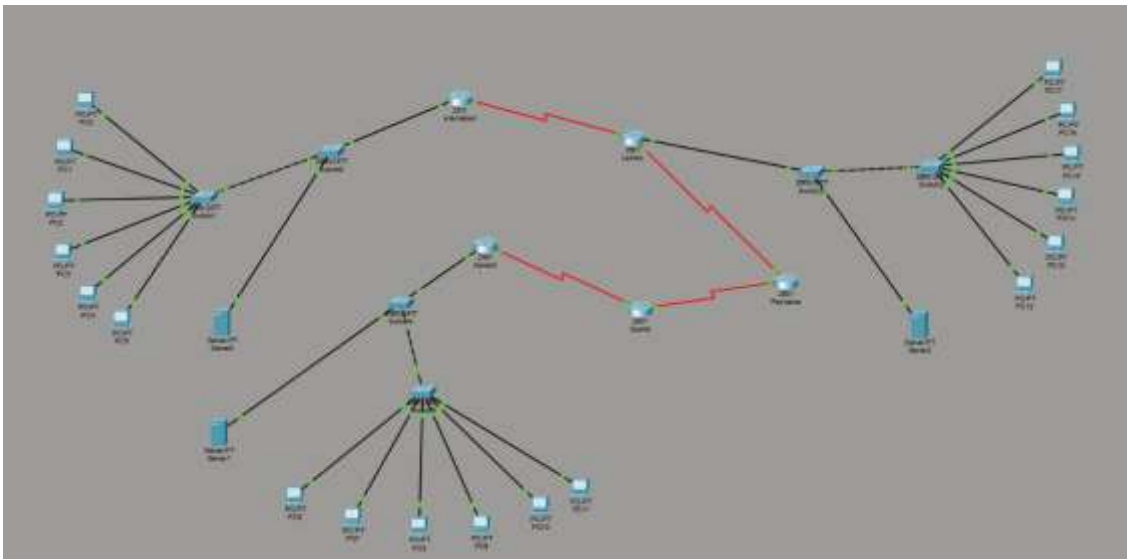
- **Islamabad Site**
- **Lahore Site**
- **Karachi Site**

Each site includes:

- One router
- Multiple switches
- Three VLANs
- Six PCs per site

- Servers for network services

The sites are interconnected through a WAN using serial links between routers.



4. LAN Configuration

4.1 VLAN Design

Each site uses **three VLANs** to separate network traffic:

VLAN ID	VLAN Name	Purpose
10	Admin	Administrative users
20	Staff	Faculty and staff
30	Students	Student users

This design improves security, performance, and traffic management.

VLAN	Name	Status	Ports
1	default	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Admin	active	Fa0/2, Fa0/3
20	Staff	active	Fa0/4, Fa0/5
30	Students	active	Fa0/6, Fa0/7
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch>

4.2 Switch Configuration

- Access ports are assigned to specific VLANs.
- Trunk ports are configured using **802.1Q encapsulation**.
- Trunks allow VLANs 10, 20, and 30.
- Spanning Tree Protocol (STP) is enabled to prevent loops.

```
Switch>show interfaces tr
Switch>show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    10

Port      Vlans allowed on trunk
Fa0/1     10,20,30

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30

Switch>
```

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.C9E2.18E4
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0001.C9E2.18E4
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Desg FWD 19       128.1    P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0001.C9E2.18E4
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
             Address     0001.C9E2.18E4
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Switch#
```

4.3 Inter-VLAN Routing

Inter-VLAN routing is implemented using **Router-on-a-Stick**:

- Sub-interfaces created on router interface G0/0
- Each sub-interface uses encapsulation dot1Q
- Default gateways are configured per VLAN

Example:

- VLAN 10 → 192.168.10.1
- VLAN 20 → 192.168.20.1
- VLAN 30 → 192.168.30.1

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset    up          up
GigabitEthernet0/0.10   192.168.10.1    YES manual   up          up
GigabitEthernet0/0.20   192.168.20.1    YES manual   up          up
GigabitEthernet0/0.30   192.168.30.1    YES manual   up          up
GigabitEthernet0/1      unassigned      YES unset    administratively down down
GigabitEthernet0/2      unassigned      YES unset    administratively down down
Serial0/3/0            10.0.0.1        YES manual   up          up
Serial0/3/1            unassigned      YES unset    administratively down down
Vlan1                unassigned      YES unset    administratively down down
Router#
```

5. WAN Configuration

5.1 Router Deployment

A total of **five routers** are used (city-wise naming):

- Islamabad Router
- Lahore Router
- Karachi Router
- (Additional routers for WAN backbone)

Each router connects to:

- LAN via Fast/Gigabit Ethernet
- WAN via Serial interfaces

5.2 PPP Encapsulation

PPP encapsulation is configured on all serial links to ensure reliable point-to-point communication.

5.3 Dynamic Routing (RIP v2)

Routing Information Protocol version 2 (RIP v2) is used to enable automatic route exchange between routers.

Features:

- Supports VLSM
- Classless routing

- Easy to configure

```
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 0 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
GigabitEthernet0/0.10 22
GigabitEthernet0/0.20 22
GigabitEthernet0/0.30 22
Serial0/3/0          22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
  192.168.10.0
  192.168.20.0
  192.168.30.0
Passive Interface(s):
Routing Information Sources:
  Gateway         Distance      Last Update
  10.0.0.2         120          00:00:04
Distance: (default is 120)
Router#

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
C       10.0.0.2/32 is directly connected, Serial0/3/0
R       10.0.1.0/30 [120/1] via 10.0.0.2, 00:00:28, Serial0/3/0
R       10.0.2.0/30 [120/2] via 10.0.0.2, 00:00:28, Serial0/3/0
R       10.0.3.0/30 [120/3] via 10.0.0.2, 00:00:28, Serial0/3/0
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
  192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
  192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
R       192.168.40.0/24 [120/1] via 10.0.0.2, 00:00:28, Serial0/3/0
R       192.168.50.0/24 [120/1] via 10.0.0.2, 00:00:28, Serial0/3/0
R       192.168.60.0/24 [120/1] via 10.0.0.2, 00:00:28, Serial0/3/0
R       192.168.70.0/24 [120/4] via 10.0.0.2, 00:00:28, Serial0/3/0
R       192.168.80.0/24 [120/4] via 10.0.0.2, 00:00:28, Serial0/3/0
R       192.168.90.0/24 [120/4] via 10.0.0.2, 00:00:28, Serial0/3/0
Router#
```

6. Network Services

6.1 DHCP Server

- DHCP is configured to automatically assign IP addresses.
- Separate pools are created for each VLAN.

- PCs receive IP address, subnet mask, and default gateway dynamically.

```
Router#show ip dhcp pool

Pool VLAN10 :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Excluded addresses                : 3
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.10.1       192.168.10.1 - 192.168.10.254  0 / 3 / 254

Pool VLAN20 :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Excluded addresses                : 3
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.20.1       192.168.20.1 - 192.168.20.254  0 / 3 / 254

Pool VLAN30 :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Excluded addresses                : 3
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.30.1       192.168.30.1 - 192.168.30.254  0 / 3 / 254
Router#
```

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

  Connection-specific DNS Suffix...:
  Link-local IPv6 Address . . . . .: FE80::20B:BEFF:FE42:1E88
  IPv6 Address . . . . . : ::
  IPv4 Address . . . . . : 192.168.10.10
  Subnet Mask . . . . . : 255.255.255.0
  Default Gateway . . . . . : ::
                               192.168.10.1

Bluetooth Connection:

  Connection-specific DNS Suffix...:
  Link-local IPv6 Address . . . . .: ::
  IPv6 Address . . . . . : ::
  IPv4 Address . . . . . : 0.0.0.0
  Subnet Mask . . . . . : 0.0.0.0
  Default Gateway . . . . . : ::
                               0.0.0.0

C:\>
```

6.2 DNS Server

- DNS server resolves device names to IP addresses.
- Routers and servers are named city-wise.
- Enables hostname-based communication.

6.3 Web Servers

- Two web servers are deployed.
- HTTP service is enabled.

- PCs can access web pages via browser.

7. Network Testing and Verification

To verify correct functionality, the following tests were performed:

7.1 LAN Testing

- Ping between PCs in the same VLAN
- Ping between different VLANs

7.2 WAN Testing

- Ping between PCs of different cities
- Traceroute to verify packet path

8. Challenges Faced

- Native VLAN mismatch on trunk ports
- Initial inter-VLAN communication failure
- Incorrect access and trunk port configuration

These issues were resolved by:

- Correct VLAN assignments
- Matching native VLANs
- Proper router sub-interface configuration

9. Conclusion

This project successfully demonstrates the design and implementation of an enterprise-level computer network. The network integrates LAN and WAN technologies, VLAN segmentation, dynamic routing, and essential network services. All project objectives were achieved, and successful communication between different sites was verified through extensive testing.

The project enhanced practical understanding of real-world networking concepts and their implementation using Cisco Packet Tracer.